

Necessary in Its Native Block: Selectivity and Gating in a Mamba-3 Block under Matched-Noise Cross-Condition Drift

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Abstract

The robustness of Mamba-style selective state-space models (SSMs) is often attributed to input-dependent selectivity. We test this attribution with a three-step ablation chain on one controlled testbed: a Mamba-3 block, bearing vibration signals (XJTU-SY, one speed-and-load transition run in both directions), and noise-aware training at matched signal-to-noise ratio. Decision thresholds for each step were written down before the corresponding grid was run; for the last grids they were first committed to a public repository. Values are macro-F1 at the final epoch of a fixed 50-epoch budget, with best-epoch values as a sensitivity analysis. First, replacing the input-dependent parts of Δ , A , B and C by learned constants costs at most about 4 pp under deep noise (paired intervals include zero). Freezing the remaining input-dependent terms of the Mamba-3 scan as well, so that no tensor entering the scan depends on the input, leaves an advantage over a parameter-matched time-invariant S4D of +21.9, +24.0 and +21.2 pp at 0, -2 and -6 dB (5/5 seeds, intervals excluding zero), with fewer parameters. Second, removing the multiplicative SiLU gate from the frozen block costs 9–23 pp and returns it to roughly the level of S4D. Third, replacing the gate in place by an additive branch with identical parameters and initial weights rescues none of that loss: the additive and gateless arms coincide at every level in both directions (differences within ± 0.6 pp, all intervals including zero; rescue ratios 0.02 and 0.01, pre-specified verdict: not rescued). Adding a gate to S4D, by contrast, recovers no more than an additive branch of the

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same size on the original transition (30% versus 41% of the gap) and only a small, direction-dependent increment on the reverse transition. The input-independent scan’s advantage and the cost of gate removal replicate on the reverse transition (+22 to +26 pp and 18–24 pp). Within this block the gate is necessary as a multiplicative operator, and what a time-invariant host gains from an added gate is largely the added structure. A separate per-class grid shows the gate-related differences concentrating on the minority inner-race class. On a second dataset (Paderborn real-damage bearings) the advantage over S4D and the cost of gate removal replicate when training and evaluation share bearings, but a pre-specified replication with evaluation bearings disjoint from the training bearings did not reproduce them: there every examined architecture, S4D included, transferred close to the majority-class level. The evidence is therefore conditional on one block, cross-condition drift with shared bearing identity (XJTU-SY in both directions and Paderborn), five seeds and matched-noise training.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet recent evidence questions it: a systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds the *simpler, time-invariant* S4D consistently ahead of Mamba-family variants in accuracy and efficiency [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across

19 all 865 cells (hash-verified), and the block can be dissected.

20 Our approach is deliberately *attributional* rather than *methodological*: we
 21 propose no new architecture and claim no state-of-the-art. Instead we take
 22 a regime in which a Mamba-style block exhibits a large, stable robustness
 23 advantage over a time-invariant SSM, and ask which ingredient earns it.
 24 The chain has three steps, each with adjudication thresholds fixed before the
 25 corresponding grid was run:

- 26 1. **Test the classical selectivity terms (necessity test).** We con-
 27 struct a *frozen* arm: the input-dependent parts of Δ , A , B and C
 28 are replaced by learned constants. The Mamba-3 scan also takes an
 29 input-dependent trapezoidal weight and input-dependent rotation an-
 30 gles; these, the convolutional stem, the multiplicative SiLU gate, the s-
 31 tate parameterisation, normalisation and projections are retained (Sec-
 32 tion 3.2). If these four terms drive robustness, the frozen arm should
 33 collapse toward the time-invariant baseline.
- 34 2. **Localise the necessary component.** On the frozen base, we remove
 35 one component at a time (convolutional stem; SiLU gate; heavy-tailed
 36 state parameterisation) under a single-variable discipline checked key-
 37 by-key on the state dictionary (Section 3.2), and adjudicate each arm
 38 against pre-specified Δ -thresholds.
- 39 3. **Test sufficiency by constructive grafting.** The component iden-
 40 tified as necessary is added to plain S4D as a gate branch, and a pre-
 41 specified recovery ratio decides whether “removal collapses” is matched
 42 by “addition restores”.

43 The outcome is a conditional pattern (final-epoch reporting; best-epoch
 44 values in Section 4.9). Freezing the input-dependent Δ , A , B , C projections
 45 costs at most about 4 pp in deep noise (paired intervals include zero) while the
 46 block retains a 15–20 pp advantage over S4D: under the pre-specified criterion
 47 these projections are not the primary contributor. Freezing the trapezoidal
 48 weight and the rotation angles as well—so that nothing entering the scan
 49 depends on the input—does not shrink the advantage either: it measures
 50 +21 to +24 pp over S4D with fewer parameters (Section 4.2). Removing
 51 the SiLU gate costs 9–23 pp and takes the block to roughly the level of
 52 plain S4D. Adding a gate branch to S4D recovers only $R = 0.20$ of the
 53 gap (0.17 under best-epoch reporting). Rebuilt in the donor’s width and
 54 projection structure, the graft recovers $R = 0.30$ (interval 0.16–0.52), but

an additive branch with identical parameters recovers at least as much (0.41, Section 4.5). The converse test closes the argument: replacing the gate inside the Mamba-3 block by an additive branch with identical parameters and initial weights rescues none of the loss on either transition (rescue ratios 0.02 and 0.01, Section 4.6). The gate is necessary as a multiplicative operator in its native block; in the S4D host it adds nothing beyond an additive branch of the same size on the original transition and a small increment on the reverse one. The first two steps replicate on the reverse transition (Section 4.7) and on a second dataset when bearings are shared between training and evaluation, but not when the evaluated bearings are new, where no examined architecture transfers (Section 4.8). A per-class analysis on a separate, shorter grid indicates that the gate-related differences concentrate on the minority class.

Contributions. (i) A layered freezing experiment on a Mamba-3 block under distribution drift and matched noise, from the classical selectivity terms (Δ , A , B , C) through to a scan in which no tensor depends on the input, in which the block’s deep-noise advantage over S4D survives every step; each arm is verified input-independent by capturing the tensors the scan receives for two different inputs. (ii) Component-level evidence that the output gate is the one ablated component whose removal causes a large, one-signed loss, and that adding it to a time-invariant host, even in the donor’s width and with an additive control of identical size, recovers no more than the added structure does, while replacing it in its own block by an additive branch of identical size recovers nothing on either transition; the first two findings also replicate on the reverse transition. (iii) Final-epoch reporting under a fixed budget for every arm of the chain, with best-epoch selection on the evaluation condition as a sensitivity analysis: absolute levels shift by 0.6–10.6 pp while no contrast changes sign. (iv) A documented pre-specification and review trail—evaluation-label hashes, thresholds written before each grid, and agent-assisted review—including the incidents in which automated adjudication departed from the pre-specified criteria; we release the trail.

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [6] evolve a linear state with input-independent parameters (A , B , C , Δ), acting as a bank of learned linear

91 filters. Mamba-style blocks [1] make Δ , B , C functions of the input token
 92 (selectivity), wrap the scan in a block containing a short causal convolu-
 93 tional stem, a multiplicative SiLU gate on a parallel branch, and residu-
 94 al/normalisation structure; the concrete block used throughout this paper
 95 follows the Mamba-3 formulation [7]. The narrative crediting selectivity for
 96 robustness draws on theory showing input-dependent dynamics can represent
 97 discontinuity-sensitive projections and counteract memory decay [8], and on
 98 [9]. We treat these as *hypotheses to be tested*, not established attribution-
 99 s: the theoretical results concern representability, which does not by itself
 100 determine which component earns empirical robustness in a trained block.

101 2.2. The unresolved comparison

102 Empirically the picture is contested. A systematic comparison on 59 clean
 103 multivariate time-series classification benchmarks (MONSTER and UEA)
 104 finds S4D-based models superior to Mamba variants [4]; that study does not
 105 include additive noise or a shift of operating condition between training and
 106 evaluation, which is the regime examined here; theoretical work identifies
 107 long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, appli-
 108 cation literatures—including machine-condition monitoring—contain many
 109 Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but
 110 none, to our knowledge, isolates *which component* of the block is responsible
 111 under controlled conditions. Our clean-data results agree with the S4D-
 112 superiority finding (S4D matches or slightly exceeds the Mamba-style block
 113 on clean data). Whether the disagreement in the literature reflects a regime
 114 difference is not settled by a single testbed; what we can report is that, in the
 115 drift $\times$ noise regime studied here, the Mamba-style block’s advantage over
 116 S4D is not removed by freezing the classical selectivity terms.

117 2.3. Testbed rationale

118 We use bearing vibration diagnosis under cross-condition drift (speed
 119 and load both change) as the testbed, for three properties that matter to
 120 the attribution question rather than to the application: (a) the distribution
 121 shift is physically generated (a change of rotation speed and radial load be-
 122 tween recorded conditions), not simulated; (b) fault signatures are sparse
 123 transients—precisely the discontinuous structures selectivity theory predicts
 124 selective SSMs should excel at [8], which makes it a plausible arena for the
 125 selectivity hypothesis; and (c) the robustness gap between the Mamba-style

block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.10). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.10 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every reported cell carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication. The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), and four follow-up grids add 255 cells (Sections 4.2, 4.9, 4.5, 4.6 and 4.7).

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals reported here captures bearing-level uncertainty. One follow-up grid (Section 4.7) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the same eight bearings with their roles swapped, so it tests the other direction

rather than independent bearings; its evaluation set holds 4,624 OR and 624 IR windows, all IR windows coming from Bearing2_1, and its label hash is 93424834.... The 35 Hz/12 kN condition has no inner-race bearing and cannot form a binary pair.

The principal arms are summarised in Table 1.

3.2. What the frozen arm freezes, and what the checks verify

In the Mamba-3 layer used here, one input projection emits the gate and value streams, the scan inputs B and C , the step-size and decay terms (Δ and A , through input-dependent additive terms), a trapezoidal weight and rotation angles. The frozen arm removes the input-dependent parts of Δ , A , B and C and replaces them with learned constants broadcast over time (for Δ the static bias of the original layer is kept; for A a new constant parameter plays the role of a static log-decay, passed through the layer’s own heavy-tailed clamp). The removed projections are deleted, so no parameter or gradient can leak. **The trapezoidal weight and the rotation angles remain functions of the current input**, as do the stem and the output gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3 formulation [7] the input-dependent rotation and trapezoidal weighting also enter the state update. Section 4.1 is accordingly a statement about the classical selectivity terms Δ , A , B , C only. Three further arms close that gap by the same device: *frozen-trap* replaces the trapezoidal weight by a learned per-head constant, *frozen-rot* replaces the rotation angles by learned constants (so the rotation becomes a learned position-only phase rather than a function of the current input), and *frozen-all* does both, leaving **no tensor entering the scan dependent on the input**. In each case the corresponding `in_proj` output slice is deleted. Section 4.2 reports them. For these arms the construction check adds the decisive probe: the tensors handed to the scan are captured for two different inputs through the same module, and a tensor the arm claims to have frozen is required to be bitwise identical across them while a tensor left selective is required to differ (all three arms pass). No kernel modification is needed, because the scan interface accepts these tensors regardless of their origin.

Single-variable discipline was checked by key-level state-dictionary diffs between two models built under the same seed. For the frozen arm the diff verifies that every non-target tensor has the same name and shape as in BM3; for the component-ablation and graft arms it additionally compares the values of non-target tensors at initialisation. Every differing key is mapped

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: no tensor entering the scan depends on the input	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Section 4.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$; same modules ⁸ parameters and initial weights (Section 4.6)	112,410
CNN/TCN baselines	Conventional baselines for context	—

199 to an entry in an architecture map. We do not claim byte-level identity of
 200 non-target tensors in the frozen arm, and trained weights naturally differ
 201 between arms; the checks concern how the arms were constructed.

202 3.3. Reporting rule

203 Every cell is trained for a fixed budget of 50 epochs with a cosine learning-
 204 rate schedule that anneals to zero, and the macro-F1 reported in the main
 205 text is that of the *final epoch*, a rule that uses no evaluation data to choose
 206 a model. The original grids recorded instead the maximum over epochs of
 207 macro-F1 on the evaluation split (the harness defines no separate validation
 208 split), and the thresholds pre-specified for those grids were written against
 209 that rule. Every arm of the chain was therefore re-run on the adjudication
 210 band (0, −2, −6 dB) with the original training code, the same evaluation
 211 fingerprint and per-epoch evaluation (Section 4.9). Adjudications are re-
 212 ported under the rule they were specified for and under the final-epoch rule,
 213 and best-epoch values are given as a sensitivity analysis. Levels outside the
 214 band (clean, +10, +6, −10 dB) and the per-class grid were not re-run; values
 215 there are best-epoch and labelled auxiliary. Because the final epoch of the
 216 same evaluation condition has been inspected repeatedly during this study,
 217 final-epoch values are not a fresh confirmatory test either.

218 3.4. Confound elimination

219 Before component attribution, three candidate confounds were examined
 220 on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D
 221 (S4D-wide − S4D: −0.4, −1.0, +0.4 pp at 0, −2, −6 dB under final-epoch
 222 reporting, all intervals including zero), while BM3-frozen exceeds S4D-wide
 223 by +15.9, +20.8, +16.0 pp with intervals excluding zero at all three levels, so
 224 this particular increase in S4D’s parameter count does not explain the gap;
 225 the control does not address the change in channel mixing that gate removal
 226 also entails. **Normalisation:** both SSM arms use LayerNorm and contain
 227 no BatchNorm; the arm that degrades under noise is the one *without* any
 228 batch statistics, which argues against BatchNorm statistics mismatch as the
 229 explanation. Other normalisation inside the Mamba-3 layer was not varied.
 230 **Auxiliary physics loss:** an exploratory kinematics-alignment regulariser
 231 carried by the original model contributes −0.1 pp pointwise (curves coincide);
 232 all attribution arms are therefore built on the pure cross-entropy variant.

233 3.5. *Pre-specification and adjudication*

234 Every adjudicated question in this study had numeric decision thresh-
235 olds written to a task card and to disk before the corresponding grid was
236 launched. This is internal pre-specification, not external registration: the
237 supporting evidence is local file modification times and run-directory names
238 (released, [Appendix A](#)), which are not tamper-evident and are not pre-
239 served by a repository checkout, so the trail lets readers inspect the order
240 of events but cannot prove it. The exception is the last three follow-up
241 grids (Sections 4.5–4.9): their thresholds, and for the last two grids the ti-
242 tle decision that depends on their outcome, were committed to the public
243 repository before the first cell was run, which gives those grids an external
244 time-stamp. Later questions (component ablation, graft) were specified after
245 earlier grids had been seen, as is natural in a staged design. Adjudication
246 was mechanical: an analysis pass applies the thresholds and emits a decision
247 artifact; an independent reviewer pass re-computes aggregates, verifies finger-
248 prints row-by-row, and must cite file paths and modification times for every
249 check. The pre-specified thresholds were binding in practice: two attempted
250 post-hoc criterion substitutions and one stale-artifact false verification were
251 caught and re-adjudicated under the original rules — this is the one place
252 where pre-specification actually bit. [Appendix C](#) details the incidents, the
253 automated executor/reviewer implementation, and the human adjudication
254 gates; all scientific decisions — thresholds, rulings, and interpretations —
255 were made by the authors and are timestamped. Statistical tests (paired,
256 per-condition) are reported throughout but were never adjudication criteria.
257 A single repeat of one 2-epoch configuration in a separate process differed
258 by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; [Appendix B](#));
259 this is an observed repeat difference, not an estimated variability bound, and
260 arm-level contrasts are judged against seed-level variability (2–9 pp).

261 4. Results

262 Throughout, “deep noise” denotes the pre-specified adjudication band
263 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-
264 invariant baseline is near chance there: a constant majority-class predictor
265 scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this
266 evaluation set, against 52.3% for S4D at the best epoch). All values are
267 macro-F1 (%), mean over 5 seeds, at the final epoch of the 50-epoch budget
268 (Section 3.3); best-epoch values are given alongside or in Section 4.9, and

values outside the adjudication band are best-epoch and auxiliary. Figure 1 summarises the three-step ablation chain that structures this section.

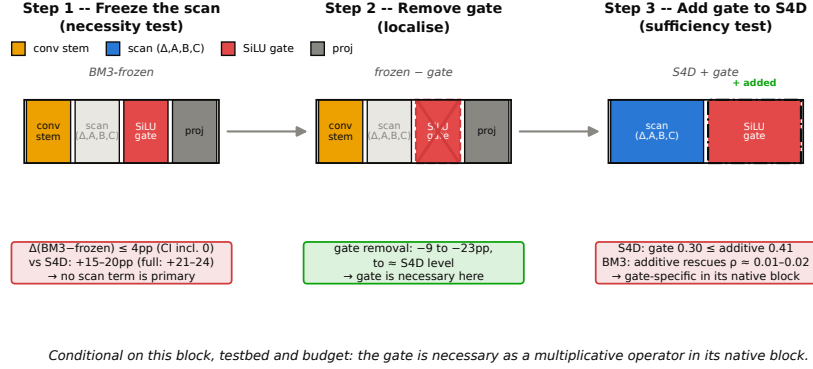


Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.3), and graft that ingredient onto S4D to test sufficiency (Section 4.4).

4.1. Freezing the input-dependent Δ, A, B, C projections

Freezing the input-dependent Δ, A, B, C projections (Section 3.2) costs at most 3.7pp anywhere in the adjudication band on the point estimates under final-epoch reporting (+3.7, +0.9, +0.2pp at 0, -2, -6 dB), while the frozen block retains an advantage of +15.5, +19.8, +16.4pp over S4D (Table 2, Figure 2). Five seeds bound these quantities loosely: paired 95% intervals for $\Delta(\text{BM3-frozen})$ are [-3.8, 11.3], [-4.5, 6.3], [-12.1, 12.6] pp (BM3 ahead in 4/5, 3/5, 2/5 seeds), and for $\Delta(\text{frozen-S4D})$ [4.3, 26.8], [12.0, 27.7], [4.1, 28.7] pp (5/5, 5/5, 4/5 seeds). The pre-specified rule attributed robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10\text{pp}$ in deep noise with the frozen arm collapsing toward S4D. It was written for best-epoch reporting, under which the original grid gives $\Delta(\text{BM3-frozen}) = +5.1, +3.3, +2.2\text{pp}$ (re-run: +5.1, +3.5, +3.0) and $\Delta(\text{frozen-S4D}) = +13.8, +17.0, +17.7\text{pp}$. Neither condition holds under either rule, so the pre-specified verdict is that these projections are *not the primary mechanism*. This is a failure to meet a confirmation criterion, not a demonstration that freezing is cheap: the intervals above are compatible with differences larger than 10pp, and no equivalence margin was pre-specified or tested. One limit applies: the frozen

Table 2: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Sections 4.2 and 4.9). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run).

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3} - \text{frozen})$	$\Delta(\text{frozen} - \text{S4D})$
<i>Final epoch (primary)</i>					
0 dB	86.9	83.1	67.6	+3.7	+15.5
−2 dB	88.3	87.3	67.5	+0.9	+19.8
−6 dB	83.3	83.1	66.7	+0.2	+16.4
<i>Best epoch (sensitivity, same re-runs)</i>					
0 dB	95.6	90.6	75.9	+5.1	+14.6
−2 dB	95.8	92.3	75.1	+3.5	+17.2
−6 dB	88.6	85.7	68.6	+3.0	+17.1
<i>Auxiliary (best epoch, original grid)</i>					
clean	78.9	78.9	81.0	−0.0	−2.1
−10 dB*	68.1	64.3	52.3	+3.8	+12.0

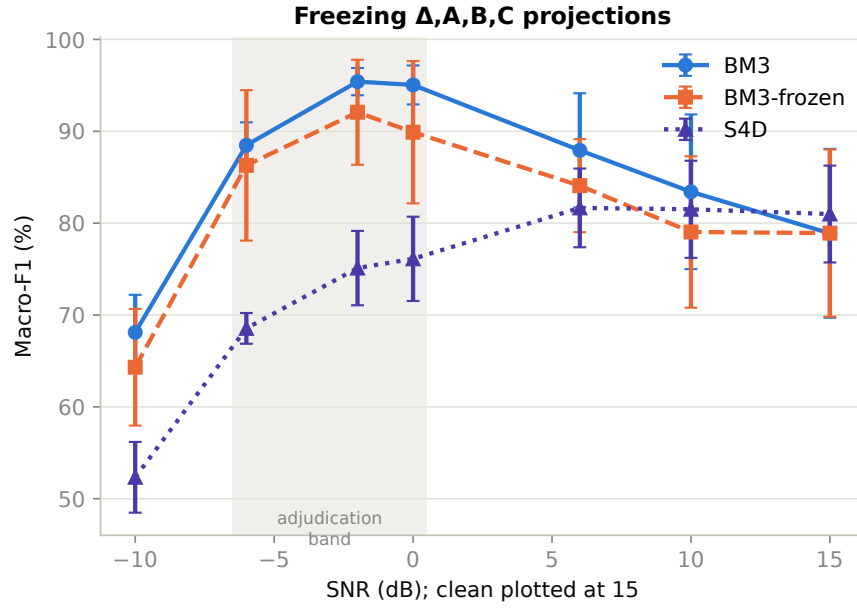


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table 2, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

arm still contains input-dependent trapezoidal weights and rotation angles, so this result concerns Δ, A, B, C only; Section 4.2 removes that limit. On clean data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-style arms—in line with the clean-benchmark comparison of [4], which, like this auxiliary row, involves no added noise. In the noisy regime, the block’s advantage over S4D is not removed by freezing these four terms, so it must be carried by other parts of the block or by the remaining input-dependent scan terms.

4.2. Freezing the remaining input-dependent scan terms

Table 3 reports the three layered arms of Section 3.2 on the adjudication band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training code, the same evaluation fingerprint, with the original BM3 arm re-run in the same grid). The question is whether the advantage that survived freezing Δ, A, B, C is carried by the two input-dependent terms the frozen arm retained.

Table 3: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen−trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen−rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen−all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen−all − S4D	+21.9	[16.7, 27.1]	+24.0	[21.4, 26.5]	+21.2	[17.5, 24.9]
BM3-frozen − frozen−all	−6.3	[−17.6, 4.9]	−4.1	[−12.0, 3.7]	−4.8	[−17.8, 8.2]
BM3 − frozen−all	−2.6	[−8.2, 2.9]	−3.2	[−7.0, 0.6]	−4.5	[−7.2, −1.9]

It is not. With every tensor entering the scan input-independent, *frozen−all* is +21.9, +24.0 and +21.2 pp above S4D at 0, −2 and −6 dB under final-epoch reporting, positive in 5/5 seeds at every level and with intervals excluding zero—if anything a larger margin than the partially frozen arm’s +15.5, +19.8, +16.4 pp (Table 11). None of the three freezing steps costs

accuracy: the paired differences BM3-frozen – frozen–trap, – frozen–rot and – frozen–all are negative at all three levels (between -1.8 and -6.3 pp under final-epoch reporting), i.e. the more heavily frozen arm is numerically ahead, with every interval including zero. The direction is not an artefact of capacity: *frozen–all* has 103,842 parameters against 112,410 for BM3-frozen and 177,938 for BM3. At the deepest level *frozen–all* also exceeds the full BM3 block by 4.5 pp ($[-7.2, -1.9]$ for BM3 – frozen–all, 0/5 seeds in BM3’s favour); we report this but do not build on it, since five seeds and one speed pair cannot separate a genuine advantage of a time-invariant scan from a variance effect.

Two readings of the whole step are therefore available, and the data do not separate them: input-dependence of the scan contributes nothing measurable here, or it contributes something that the learned constants recover under this training budget. Either way the conclusion for the attribution chain is the same and now rests on a complete intervention rather than a partial one: **the Mamba-style block’s deep-noise advantage over a time-invariant S4D does not require any input-dependent term in its scan**. What remains to carry it is the rest of the block—the convolutional stem, the heavy-tailed A parameterisation, the normalisations and the output gate—which is what Sections 4.3–4.4 examine. The re-run also reproduces the original grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed the original frozen-selectivity grid’s 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build, Section 4.9).

4.3. Removing the gate is the only ablation with a large, one-signed effect

“Necessary” is used here in the operational sense of this ablation: removing the gate from the frozen block, with the other components and the budget unchanged, removes most of the block’s advantage. The gate result is large, one-signed and grows with noise depth: under final-epoch reporting removal costs 9.1, 15.6, 22.6 pp at 0, -2 and -6 dB (paired 95% intervals $[2.9, 15.3]$, $[-0.8, 32.0]$, $[5.8, 39.4]$; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp threshold at every level; the interval at -2 dB includes zero because of the large seed variance of the gateless arm. Under the best-epoch rule for which the threshold was written, the component-ablation grid gives 13.1, 21.1 and 23.5 pp ($[6.5, 19.6]$, $[3.8, 38.5]$, $[8.5, 38.6]$; 5/5 seeds at every level), so the verdict is *primary* under both rules. The gateless arm gives up most of the Mamba advantage but is not resolved against plain S4D: frozen–gate – S4D

Table 4: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points: final-epoch mean, 95% t -interval over seeds and number of seeds (of 5) with a positive difference (re-runs of Section 4.9); “best” is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
SiLU gate	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
conv stem ($k=7$)	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
heavy-tailed A → S4D-style A	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; inter- vals include 0

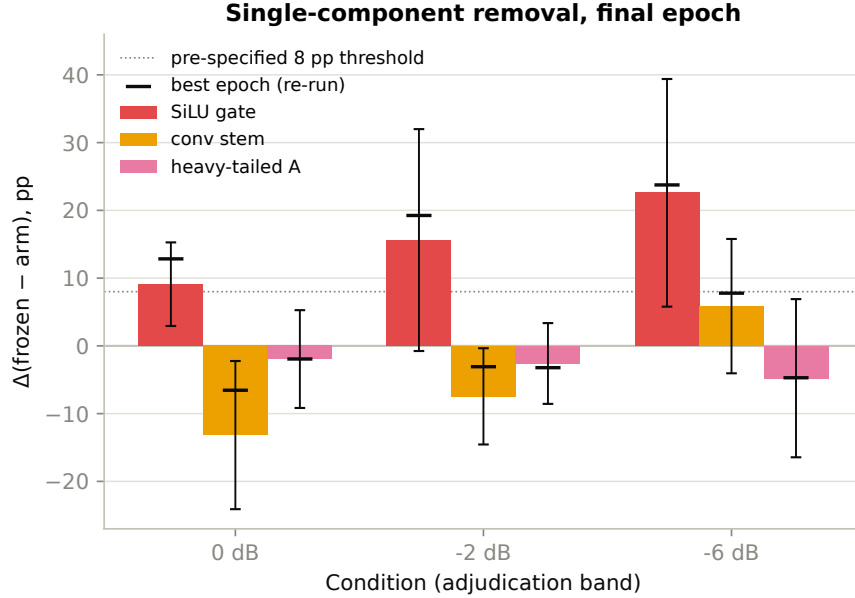


Figure 3: Component-removal Δ (percentage points, frozen - arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t -intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold.

345 is +6.4, +4.2, -6.2 pp, with every interval including zero. We therefore de-
 346 scribe it as roughly at the level of S4D, below it only on the point estimate at
 347 -6 dB; this suggests, but does not establish, that components interact rather
 348 than add. The other two components tell regime stories rather than robust-
 349 ness stories. Removing the convolutional stem is beneficial at 0 and -2 dB
 350 (frozen - frozen-conv = -13.2 [-24.1, -2.2] and -7.5 [-14.6, -0.3], with
 351 the stemless arm ahead in every seed) and costly at -6 dB (+5.9, [-4.0, 15.8],
 352 3/5 seeds); under best-epoch reporting the pattern was the same (-6.6, -2.7,
 353 +7.9 pp). On clean cross-speed data (auxiliary, best epoch, original grid) the
 354 stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing
 355 from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pat-
 356 terns mistransferring across conditions, or with optimisation friction under
 357 the matched budget; we report both readings. Replacing the heavy-tailed
 358 state parameterisation by an S4D-style one is numerically beneficial at all
 359 three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under
 360 best-epoch reporting), but every paired interval includes zero, so we draw no
 361 directional conclusion from it.

362 4.4. Adding a narrow gate branch to S4D

Table 5: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only
 the gate branch added). Upper block: final epoch, all three arms from the same re-run
 (Section 4.9). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid
 and S4D and BM3-frozen, and hence R , from the component-ablation grid of Section 4.3
 (independent runs of the same arms differ by 0.1–0.7 pp).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
<i>Final epoch (primary)</i>				
0 dB	71.5	67.6	83.1	0.249
-2 dB	74.3	67.5	87.3	0.341
-6 dB	66.6	66.7	83.1	-0.003
<i>Best epoch (as pre-specified)</i>				
0 dB	79.8	76.2	90.6	0.249
-2 dB	79.4	75.2	92.8	0.239
-6 dB	69.2	68.6	86.0	0.035

363 The pre-specified recovery ratio is $R = \text{mean over the band of (S4D+gate-}$
 364 $\text{S4D})/(\text{frozen} - \text{S4D})$. Under final-epoch reporting $R = \mathbf{0.196}$ (per level

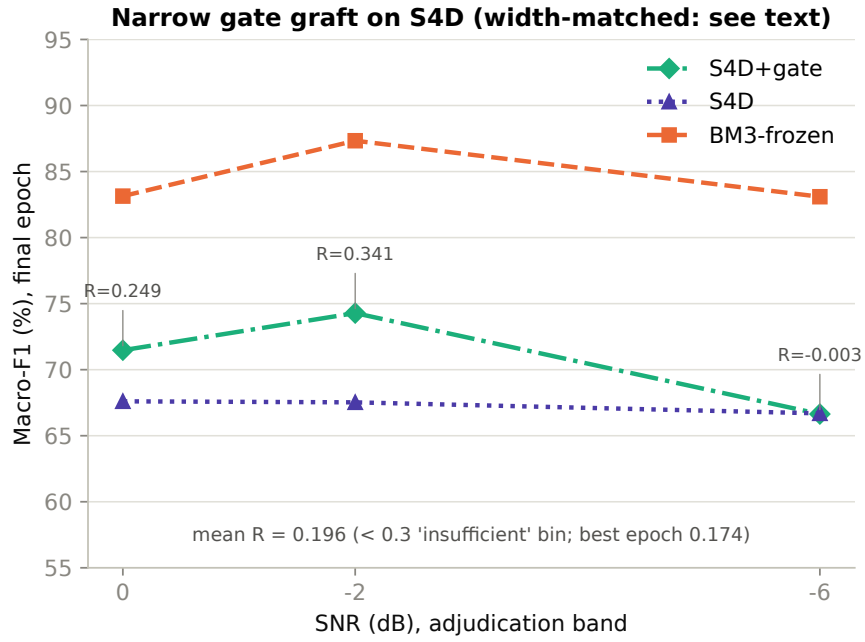


Figure 4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table 6.

0.249, 0.341, -0.003 ; seed-resampled 95% interval $[0.09, 0.31]$); under the best-epoch rule for which it was specified, $R = 0.174$ ($[0.04, 0.29]$). The pre-specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); both values fall in the last, although the final-epoch interval reaches the 0.3 boundary. The per-level paired gains of S4D+gate over S4D under final-epoch reporting are $+3.9$ ($[-0.8, 8.6]$), $+6.8$ ($[2.4, 11.1]$), -0.1 ($[-4.0, 3.9]$) pp. At the deepest adjudicated level the graft recovers nothing. Removal and addition are not mirror-image interventions. In BM3 the gate has its own $64 \rightarrow 128$ projection and multiplies the scan output at the inner width of 128 before the output projection; removing it deletes 32,768 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch (with bias) computed from the layer input and multiplying the SSM output before the residual addition; it adds 16,640 parameters over four layers. The two share the multiplicative SiLU form but not width, projection structure or parameter count. Removal costs 9–23 pp in the frozen block, whereas this graft changes S4D by between -0.1 and $+6.8$ pp (final epoch). What this shows is that *this gate branch, in this host and under this budget, recovers little*. Three readings fit the data and cannot be separated here: (i) host dependence, i.e. the gate needs the stem or scan context it co-trained with; (ii) implementation differences in width, projection and placement; (iii) transplant friction, i.e. the grafted branch is harder to optimise in a foreign host under the matched budget (training converges normally, but convergence is not robustness). Section 4.5 tests reading (ii) with a width-matched graft and an equal-capacity non-multiplicative control; a tuning or longer-budget comparison selected on a source-domain validation set was not run.

The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) - F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under last-epoch reporting $I = 5.2$ $[2.3, 8.1]$, 8.9 $[-6.2, 23.9]$ and 22.6 $[2.9, 42.4]$ pp at 0, -2 and -6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 $[1.4, 14.2]$, 14.4 $[-1.4, 30.3]$ and 23.1 $[8.7, 37.4]$ pp. A positive I means the gate adds more on the frozen host than on S4D. Because the host differs in several respects at once and the two gates are not aligned, I cannot be assigned to a specific component or interaction term, and the seed pairing across arms is nominal (arms share seed indices, not initialisations).

4.5. A width-matched graft and an additive control

The graft of Section 4.4 is narrower and smaller than the donor gate. Two further arms rebuild it in the donor’s shape (Table 1). In each layer the layer input is projected to x and z of width 128 (the frozen block’s inner width), the S4D layer runs on x at width 128 with state size 64—so that its own parameter count and total state size equal the host’s 64-wide, state-128 layer—and an output projection maps back to width 64. *S4D+gate* (*wm*) combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938), which biases the test towards recovery. A construction check with forward hooks confirms that the tensor entering the output projection equals the intended formula for each arm and not the other, and that the two arms are identical at initialisation. The thresholds were committed publicly before the grid ran (Section 3.5): R as before, under final-epoch reporting; the verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a bin boundary, in which case we write at the tier less favourable to the claim; and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an interval excluding zero.

Table 6: Width-matched graft and additive control on the original transition. Macro-F1 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm – S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and the narrow graft are the re-runs of Section 4.9.

Arm	0 dB	−2 dB	−6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				−0.112	[−0.26, 0.02]

The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159, 0.298, 0.431). Its interval, [0.16, 0.52], crosses the 0.3 boundary, so the pre-specified verdict is INCONCLUSIVE between “insufficient” and “partial”, and we report it at the less favourable tier: a graft built in the donor’s shape recovers part of the gap. The recovery is not gate-specific. The additive branch, with the same parameters, recovers at least as much ($R = 0.408$),

$\Delta R = -0.112$ $[-0.26, 0.02]$, and at 0 dB the gated arm is below the additive one by 3.0 pp ($[-4.5, -1.5]$, 0/5 seeds). With frozen—all in the denominator the gated arm’s ratio is 0.231 $[0.13, 0.33]$; under best-epoch reporting the two ratios are 0.459 and 0.483. What a graft recovers on this transition therefore follows the added structure—a wider channel axis and input/output projections—rather than the multiplicative interaction. Reading (ii) of Section 4.4 is partly confirmed, in that the narrow graft understated what added structure can recover; but in the S4D host the gate itself contributes nothing beyond an equal-capacity additive branch, whereas removing it from the frozen Mamba-3 block costs 9–23 pp under the same rule (Table 11).

4.6. Replacing the gate in its native host

The previous subsection changes the host and keeps the operator. A pre-specified grid (15 cells, original transition) does the converse: it keeps the host and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$; modules, parameter count (112,410) and initial weights are bitwise those of BM3-frozen, and a construction check confirms that the donor operator is $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$, mean over levels, final epoch, with a seed-resampled interval; *not rescued* requires $\rho < 0.3$ with an upper bound below 0.5.

The additive branch rescues nothing. *frozen+add* is indistinguishable from the gateless arm at every level (+0.6, −0.2, +0.4 pp, every interval including zero), and the multiplicative arm exceeds it by 8.5, 15.8, 22.2 pp. The rescue ratio is $\rho = 0.024$ (per level 0.070, −0.015, 0.016), with a seed-resampled 95% interval of $[-1.159, 0.486]$; under best-epoch reporting $\rho = 0.005$. By the pre-specified rule the verdict is *not rescued*. Two features of the interval should be read with it: its upper bound lies only 0.014 below the threshold, and its lower bound is wide because the 0 dB denominator is only 9.1 pp (in 7.8% of resamples one level’s denominator fell below 2 pp and was excluded, as pre-specified).

A further pre-specified grid (15 cells) repeats the substitution on the reverse transition (Section 4.7), with BM3-frozen and frozen–gate taken from the reverse grid. The outcome is the same (Table 7, lower block): *frozen+add* coincides with the gateless arm at every level (+0.3, −0.3, +0.6 pp, every interval including zero), and the multiplicative arm exceeds it by +18.1, +22.3,

Table 7: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
<i>Original transition (37.5 Hz/11 kN → 40 Hz/10 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
BM3-frozen − frozen+add	+8.5		+15.8		+22.2	
	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
frozen+add − frozen−gate	+0.6		−0.2		+0.4	
	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	
<i>Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \text{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
BM3-frozen − frozen+add	+18.1		+22.3		+23.4	
	[1.2, 35.0] (5/5)		[3.5, 41.0] (5/5)		[6.8, 40.0] (5/5)	
frozen+add − frozen−gate	+0.3		−0.3		+0.6	
	[−0.6, 1.2] (3/5)		[−1.0, 0.4] (2/5)		[−0.2, 1.4] (5/5)	

+23.4 pp (intervals excluding zero, 5/5 seeds at every level). The rescue ratio is $\rho = 0.009$ (per level 0.017, -0.014 , 0.023), with a seed-resampled 95% interval of $[-0.943, 0.362]$; under best-epoch reporting $\rho = -0.005$. The pre-specified verdict is again *not rescued*, here with the upper bound 0.14 below the threshold.

Together with Section 4.5 this gives the two halves of a symmetric test. In the S4D host the gate adds nothing beyond an additive branch of the same size; in the Mamba-3 host an additive branch of the same size adds nothing beyond removing the gate, on both transitions. The benefit of the gate is specific to the multiplicative operator *in its native host*, and what transfers to the time-invariant host is the added structure, not the gating.

4.7. The reverse transition

A pre-specified grid repeats the chain’s central contrasts on the reverse transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen–all, BM3-frozen, frozen–gate, S4D and S4D+gate (wm), at three levels and five seeds; Section 3.1 for the data). This direction is harder for the time-invariant model: on its evaluation set, predicting the majority class gives a macro-F1 of 46.8% and uniform random guessing 41.5%, and S4D sits close to those levels (Table 8).

Table 8: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm (best / final)	0 dB	−2 dB	−6 dB
frozen–all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen–gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
frozen–all − S4D	+26.3 [21.5, 31.0] (5/5)	+25.6 [21.1, 30.1] (5/5)	+22.2 [18.8, 25.6] (5/5)
BM3-frozen − frozen–gate	+18.4 [1.2, 35.6] (5/5)	+22.0 [3.4, 40.6] (5/5)	+24.0 [7.6, 40.3] (5/5)
frozen–gate − S4D	+5.0 [−17.6, 27.7] (2/5)	+2.4 [−19.2, 23.9] (2/5)	−1.8 [−18.9, 15.3] (1/5)
S4D+gate (wm) − S4D	+21.7 [15.4, 28.0] (5/5)	+21.5 [17.0, 26.0] (5/5)	+20.5 [15.7, 25.2] (5/5)

480 The first two steps replicate. The fully input-independent scan exceeds
 481 S4D by +26.3, +25.6 and +22.2 pp at 0, -2 and -6 dB, with intervals ex-
 482 cluding zero and 5/5 seeds at every level, so by the pre-specified rule (at
 483 least two of three levels) the first step is replicated on the reverse transition.
 484 Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the
 485 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds,
 486 and the gateless arm is indistinguishable from S4D (intervals include zero).
 487 On the third step, which the pre-specification treats as descriptive here, the
 488 width-matched graft recovers most of the gap, $R = 0.911$ [0.82, 1.04] (0.863
 489 with frozen—all in the denominator). A further pre-specified grid (15 cells)
 490 adds the same-size additive control in this direction: it reaches 67.8, 72.1,
 491 68.4% (final epoch) and recovers $R = 0.758$ [0.66, 0.88]. The gate arm ex-
 492 ceeds it by +5.1, +1.7, +3.8 pp (only the -6 dB interval excludes zero),
 493 and the difference in recovery is $\Delta R = 0.153$ [0.06, 0.26]; under best-epoch
 494 reporting it shrinks to 0.037 [-0.1, 0.17]. By the pre-specified descriptive
 495 labels this is *unresolved*: in this direction the gate carries a small increment
 496 over the additive branch, whereas on the original transition it carried none
 497 ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available
 498 to an additive branch of the same size, and the remaining gate increment is
 499 direction-dependent. Both ratios rest on an S4D baseline close to the trivial
 500 floor (51–52% against 46.8% for the majority class). Both statements carry
 501 the limits of this pair: the same eight bearings with their roles swapped, and
 502 a single inner-race bearing in the evaluation set.

503 4.8. External validity: two settings on a second dataset

504 A pre-specified grid (75 cells; thresholds, a ceiling-and-floor check and the
 505 writing consequences committed before the first cell) repeats the three central
 506 contrasts on the Paderborn KAt data [16], using real-damage bearings only.
 507 Models train on two outer-race (KA04, KA16) and two inner-race bearings
 508 (KI04, KI14) at 1500 rpm and are evaluated on five different bearings (KA15,
 509 KA22; KI17, KI18, KI21) at 900 rpm, with load and radial force unchanged;
 510 the vibration channel is resampled to 25.6 kHz so that a window spans 80 ms
 511 as in XJTU-SY, and everything else (noise, training, arms, budget) is as in
 512 the main grids. The pre-specified smoke check (two arms, one extra seed)
 513 passed its three criteria, the learnability criterion by 0.3 pp. A second pre-
 514 specified grid (45 cells, three arms) then changed one factor only: training
 515 and evaluation use the same four bearings, at the same two speeds, from
 516 different recordings.

Table 9: Paderborn, real-damage bearings, 1500 rpm $\rightarrow$ 900 rpm. *Disjoint*: training on KA04, KA16, KI04, KI14 and evaluation on KA15, KA22, KI17, KI18, KI21. *Shared*: training and evaluation on KA04, KA16, KI04, KI14 (different recordings). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds (majority-class macro-F1: 37.5% disjoint, 33.3% shared). Lower block: paired differences under final-epoch reporting, 95% t -interval and seeds (of 5) with a positive difference. “—”: arm not run in that setting.

Arm / contrast	Disjoint bearings			Shared bearings		
	0 dB	−2 dB	−6 dB	0 dB	−2 dB	−6 dB
BM3-frozen	51.9	50.3	47.7	84.5	85.8	86.6
frozen—all	54.2	51.3	47.8	—	—	—
S4D	55.6	53.9	51.1	76.1	77.3	77.9
frozen—gate	56.9	54.9	51.1	75.5	78.1	81.3
frozen+add	56.9	54.8	51.0	—	—	—
	−3.8	−3.6	−3.4	+8.4	+8.5	+8.7
BM3-frozen	[−11.9, 4.3]	[−12.6, 5.4]	[−8.9, 2.1]	[−1.3, 18.1]	[0.2, 16.8]	[4.3, 13.0]
— S4D	(1/5)	(2/5)	(1/5)	(4/5)	(5/5)	(5/5)
	−5.0	−4.6	−3.3	+9.0	+7.7	+5.3
BM3-frozen	[−8.8, −1.3]	[−9.2, 0.1]	[−7.1, 0.4]	[5.2, 12.8]	[5.0, 10.4]	[2.4, 8.2]
— frozen—gate	(0/5)	(0/5)	(0/5)	(5/5)	(5/5)	(5/5)
	−1.4	−2.5	−3.3			
frozen—all	[−4.0, 1.2]	[−5.3, 0.2]	[−5.8, −0.8]			
— S4D	(1/5)	(1/5)	(0/5)	—	—	—
	0.0	0.0	−0.1			
frozen+add	[−0.9, 0.8]	[−0.4, 0.3]	[−0.4, 0.2]			
— frozen—gate	(2/5)	(3/5)	(1/5)	—	—	—

517 None of the pre-specified contrasts replicates (Table 9). The fully input-
518 independent block does not exceed S4D (frozen-all – S4D: $-1.4, -2.5,$
519 -3.3 pp, the interval at -6 dB excluding zero in the negative direction), nor
520 does BM3-frozen ($-3.8, -3.6, -3.4$ pp). Removing the gate does not cost ac-
521 curacy: BM3-frozen – frozen-gate is $-5.0, -4.6, -3.3$ pp, below zero in ev-
522 ery seed at every level. With no gate effect to rescue, the native-substitution
523 test is not assessable; the additive and gateless arms again coincide ($0.0, 0.0,$
524 -0.1 pp). The pre-specified labels are therefore *not replicated* for both freeze
525 contrasts and for gate removal, and *not assessable* for the substitution test.

526 Two observations bound how this result can be read, and neither changes
527 the labels. First, the testbed discriminates poorly: every arm ends between
528 47.7 and 56.9% macro-F1 at every level, all arms fit the training condition
529 (the gated arms reach the lowest training loss), and none transfers far above
530 the level reached after the first epoch. Second, where differences are resolved
531 they run against the Mamba-3 arms, which reach lower training loss and
532 transfer slightly worse. In the disjoint setting, therefore, the block’s advan-
533 tage under matched noise and drift is not observed; we do not know whether
534 this reflects a property of the PU transition (speed reduction to 900 rpm with
535 disjoint bearings), the protocol transferred unchanged from XJTU-SY, or a
536 limit of the XJTU-SY result itself, which the shared-bearing setting below
537 separates.

538 *Shared bearings.* With bearing identity shared between training and e-
539 valuation, the same condition pair becomes transferable: every arm ends
540 between 75.5 and 86.6% at every level, and the pre-specified transferability
541 gate (5-seed mean $\geq 65\%$ with one-sided 95% lower bound $> 60\%$ at two
542 of three levels for at least one arm) is passed by all three arms at all three
543 levels (smallest lower bound 71.3%). Within this setting both effects repli-
544 cate under the pre-specified rule (Table 9, right). BM3-frozen exceeds S4D
545 by $+8.4, +8.5, +8.7$ pp at $0, -2, -6$ dB, with intervals excluding zero at -2
546 and -6 dB; removing the gate costs $+9.0, +7.7, +5.3$ pp, with intervals ex-
547 cluding zero at all three levels and in every seed; no level runs in the opposite
548 direction. The gateless arm returns to the level of S4D (frozen-gate – S4D:
549 $-0.6, +0.8, +3.4$ pp, all intervals including zero), as on XJTU-SY. Under
550 best-epoch reporting the freeze contrast is resolved at all three levels ($+6.9,$
551 $+10.8, +13.7$ pp) and gate removal only at -6 dB ($+1.5, +3.7, +6.4$ pp). The
552 cost of gate removal reaches the 8 pp threshold used on XJTU-SY at one level
553 only; we compare directions, not magnitudes, across datasets.

554 Read together, the two settings separate the factors. Moving from disjoint

to shared bearings raises every arm, S4D included, by 18.6–38.8 pp (independent seeds; per-arm Welch intervals all exclude zero), so bearing non-overlap breaks transfer for all three examined architectures rather than for the gated block alone, and once transfer is possible the block’s advantage and the cost of gate removal reappear. One property of the shared setting bounds what this shows: with two bearings per class, a bearing’s identity determines its label, so a model can succeed partly by recognising bearing-specific signatures across conditions. The same holds for every XJTU-SY transition, which also share bearings. Our results therefore concern robustness to cross-condition drift when bearing identity is shared between training and evaluation; they do not show robustness when the evaluated bearings are new.

4.9. Sensitivity to the reporting rule

The main tables use the final epoch of the 50-epoch budget (Section 3.3). The original grids recorded instead the best epoch on the evaluation condition, and their pre-specified thresholds were written for that rule. To measure how much the rule shapes the conclusions, every arm of the chain was re-run over the adjudication band with per-epoch evaluation, using the original training code and the same evaluation fingerprint: the four arms BM3-frozen, S4D, frozen–gate and S4D+gate (60 cells), the BM3 arm alongside the layered freezes (Section 4.2), and the capacity control and the two other single-component removals, S4D-wide, frozen–conv and frozen–A (45 cells, pre-specified with a rule for changes of sign). The re-runs used a different PyTorch/Triton build from the original grids; their best-epoch values reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp across the eight re-run arms (the largest deviation is in the high-variance frozen–gate arm at -2 dB).

Best-epoch values exceed final-epoch values by 0.6–10.6 pp across the 24 arm-by-level means, so best-epoch absolute levels are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0, -2 , -6 dB), and the contrasts that carry the argument are correspondingly stable (Table 11).

Selectivity. The frozen block’s advantage over S4D is $+15.5$, $+19.8$ and $+16.4$ pp (and $+21.9$, $+24.0$, $+21.2$ pp for the fully frozen scan, Section 4.2), versus $+14.6$, $+17.2$ and $+17.1$ pp under best-epoch reporting in the re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified

Table 10: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen−gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen−conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen−A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Table 11: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

threshold at every level; the interval at -2 dB includes zero ($[-0.8, 32.0]$ pp, 4/5 seeds) because of the large seed variance of the gateless arm (standard deviation 14.6 pp at -2 dB). *Gateless versus S4D*. Under final-epoch reporting the gateless arm is not below S4D at 0 and -2 dB (point estimates $+6.4$ and $+4.2$ pp) and is below it at -6 dB (-6.2 pp); all three intervals include zero. The statement that gate removal “falls below S4D” is therefore supported only as a point-estimate observation at the deepest level, and we phrase it as “to or below” elsewhere. *Graft*. The recovery ratio is 0.249, 0.341 and -0.003 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the pre-specified “insufficient” bin on the point estimate. The seed-resampled 95% interval is $[0.09, 0.31]$ and so reaches the 0.3 boundary: with five seeds the data support “far below confirmation (0.7) and probably below 0.3”, not a precise value of R . The graft gain over S4D is resolved at -2 dB only ($+6.8$ pp, $[2.4, 11.1]$) and is indistinguishable from zero at -6 dB.

Remaining arms. By the rule pre-specified for the 45-cell grid, no arm changes sign between the two rules at any level where both values exceed 1 pp, so all main-table entries are reported under the final-epoch rule with unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-wide by $+15.9$, $+20.8$, $+16.0$ pp (intervals excluding zero at all three levels), and S4D-wide $-$ S4D is -0.4 , -1.0 , $+0.4$ pp (intervals including zero). For the stem the final-epoch rule sharpens the regime trade-off (intervals at 0 and -2 dB now exclude zero) without reversing it.

Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, last-epoch values are also not a fresh confirmatory test. The reporting rule therefore changes absolute levels and the strength of the gateless-versus-S4D statement, but not the ordering BM3-frozen $>$ {S4D+gate, frozen-gate, S4D}, the size of the frozen block’s advantage over S4D, or the verdicts of the pre-specified gate and graft rules.

4.10. The interaction effect concentrates on the hard class

Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch selection on the evaluation condition: 5 arms $\times$ {clean, 0, -6 dB} $\times$ 5 seeds, per-sample predictions dumped and re-verified against reported metrics cell-by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate presence: in all three gated arms IR recall at -6 dB exceeds its clean value (e.g., $0.55 \rightarrow 0.81$ for BM3), while in both gateless arms it is below its clean value (frozen-gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone,

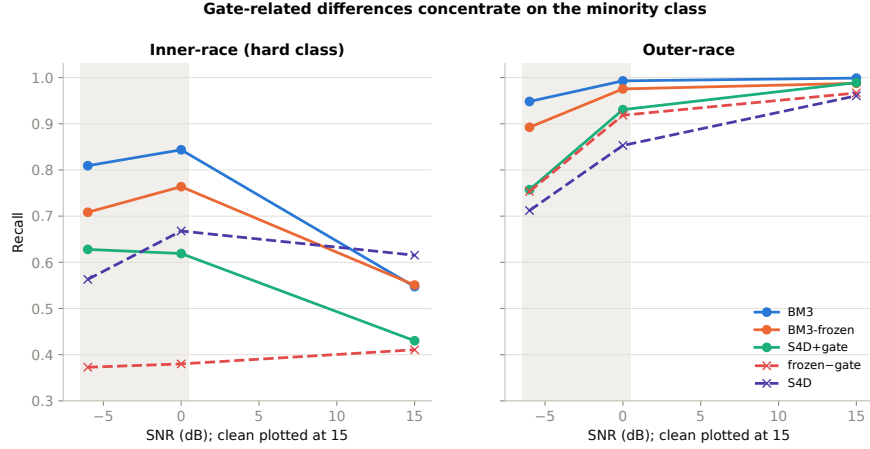


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

629 rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—
 630 gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from
 631 predictions pooled over seeds), and the graft’s noise-time gains also concen-
 632 trate on IR (−6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages
 633 the two classes, the arm-level robustness gaps of Sections 4.1–4.4 are carried
 634 substantially by the hard class. The gate-related differences are therefore
 635 class-concentrated rather than uniform, and fall on the class with the lowest
 636 recall. We did not measure classification margins, so we do not describe IR as
 637 a “low-margin” class. Because this grid is separate from the main 50-epoch
 638 grid and the IR class comes from a single training bearing (Section 3.1), these
 639 results are descriptive and do not decompose the main-grid effects.

640 5. Discussion

641 **What can be credited?** The ablation chain supports a narrow, con-
 642 ditional statement: within this Mamba-3 block, on this testbed and with
 643 noise-aware training, (i) making the scan fully input-independent—first the
 644 Δ , A , B , C projections, then the trapezoidal weight and rotation angles as
 645 well—did not remove the block’s deep-noise advantage over a time-invariant
 646 S4D, and at the deepest level the fully frozen arm matched or exceeded the
 647 original block with fewer parameters, (ii) removing the output gate removed

648 most of that advantage—(i) and (ii) holding on the reverse transition as
 649 well—and (iii) adding the gate to S4D recovered part of the advantage only
 650 together with added width and projections, and an additive branch of the
 651 same size recovered as much, while (iv) in the Mamba-3 block an additive
 652 branch of the same size recovered none of what removing the gate cost, on
 653 both transitions. These observations are consistent with a gate benefit that is
 654 specific to the multiplicative operator in its native host. They do not identify
 655 which other component the gate depends on. Nor do they separate “input-
 656 dependence contributes nothing here” from “what it contributes is recovered
 657 by learned constants under this budget”; the two readings are indistinguish-
 658 able in our data and lead to the same attribution. Representability results for
 659 selectivity [8] are not contradicted by a single ablation on one testbed. The
 660 S4D-superiority finding [4] was obtained on clean classification benchmarks,
 661 without matched noise or a shift of operating condition, so it does not speak
 662 to the regime studied here and we do not read it as covering it. Two points
 663 of contact are nevertheless consistent with it: our clean-data parity between
 664 S4D and the Mamba-style arms, and our finding that input-dependent se-
 665 lectivity is not needed for the block’s advantage. What this paper adds is
 666 specific to the matched-noise, cross-condition regime: there S4D trails the
 667 frozen Mamba-3 block by 15–26 pp under final-epoch reporting (partial and
 668 full freeze, both transitions), and the gap is attributable to the multiplicative
 669 output gate in its native block rather than to selectivity. This is a statement
 670 about a different regime, not a contradiction of the clean-benchmark result.

671 **Implications for backbone selection and for transplant practice.**
 672 For practitioners choosing backbones for noisy, drifting signal regimes, the
 673 results suggest that the block’s benefit should be judged as a composition,
 674 and that a gate added to a different host should not be expected to reproduce
 675 it: in our S4D host a gate built in the donor’s shape recovered no more than
 676 an additive branch of the same size on the original transition (Section 4.5)
 677 and only slightly more on the reverse one (Section 4.7), and neither design
 678 was tuned. A hypothesis for future work follows: a recomposed block guided
 679 by the component map (no conv stem, gate retained, S4D-style A) may match
 680 or exceed the original on this testbed; we leave it untested, because post-hoc
 681 recomposition after seeing the map would not carry the evidential status of
 682 the pre-specified chain.

683 **Why might gating be composition-bound?** We refrain from a mech-
 684 anistic claim beyond the data, but note three compatible observations. First,
 685 the gate-related differences concentrate on the minority, lowest-recall class

(Section 4.10), and matched-noise training lifts that class *only when the gate is present*. One hypothesis, which we did not test, is that the gate is the component through which matched-noise training improves hard-class recall, i.e., that it exploits noise-aware training rather than acting as a test-time denoiser; we measured neither classification margins nor gate activations, and the reading is in any case bounded by the matched-SNR scope (Section 6). Second, the multiplicative form admits a per-sample suppression interpretation, but no gate statistics were recorded, so this remains a hypothesis rather than an observation. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other components being calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them); this is the weakest of the three observations, since the gateless-S4D gap is not resolved with five seeds (Section 4.9). All are testable in future work via pairwise interaction grids and mismatched-noise protocols.

6. Boundaries and Limitations

We state the scope plainly.

1. **Testbed scope.** All claims are bounded to bearing vibration from one pair of operating conditions of one dataset (37.5 Hz/11 kN and 40 Hz/10 kN; speed and load both change), binary classification, GPU execution (no edge/real-time claims). The pair is run in both directions, but the reverse direction uses the same eight bearings with their roles swapped and has a single IR bearing in its evaluation set, so it is not an independent replication; the third condition of the dataset has no IR bearing. The five seeds do not resample bearings (Section 3.1). Independent bearings, multi-class settings and a second dataset are future work.
2. **Attribution level.** Credit assigned to “the rest of the composition” is by elimination within the enumerated component set; pairwise interaction terms are not decomposed, and the host-by-gate contrast of Section 4.4 is a four-arm estimate that cannot be assigned to a specific component.
3. **The transplant conclusion is budget-conditional.** All arms share a matched 50-epoch budget; the grafted gate trains normally yet recovers little, and convergence is not robustness (Section 4.4)—but we

cannot exclude that a longer or graft-tuned budget would recover more. Our statement is therefore limited to *this gate branch, in this host, under a matched training budget*.

4. **Matched-SNR scope.** Training and evaluation share the same injected SNR throughout (Section 3.1); “robustness” in this paper is matched-noise robustness under noise-aware training, not clean-train/noisy-test generalisation. The gate’s benefit may operate by exploiting matched-noise training (consistent with Section 4.10) rather than by test-time denoising; attribution under mismatched-noise protocols is untested and may differ.
5. **Graft dual reading.** Interaction dependence vs. transplant friction (Section 4.4), the latter now subsumed under the budget condition of item 3.
6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a TCN baseline slightly overtakes the Mamba-style block, and -10 dB is near-chance for the time-invariant baseline and excluded from adjudication; the demonstrated advantage regime is matched, Gaussian-dominated noise.
7. **Conv-stem dual reading.** Its clean-data cost admits both representation-mistransfer and optimisation-difficulty readings, reported in place (Section 4.3).
8. **Reporting rule.** Reported values are final-epoch macro-F1 under a fixed 50-epoch budget. The original grids’ best-epoch rule, for which the thresholds of the early grids were pre-specified, is reported as a sensitivity analysis and is optimistic by 0.6–10.6 pp (Section 4.9). Every arm of the chain was re-run on the three adjudication levels; the $+10/+6$ /clean/ -10 dB levels and the per-class grid were not re-run and are auxiliary (best epoch). No source-domain validation rule was available, and because the final epoch of the same evaluation condition has been inspected repeatedly, final-epoch values are not a fresh confirmatory test.
9. **Scope of the freezing intervention.** The chain’s first step freezes the input-dependent parts of Δ , A , B , C only; the fully input-independent scan is a separate grid (Section 4.2) on the three adjudication levels, which was specified after the earlier grids had been seen. No equivalence margin was pre-specified for the cost of freezing, so “no measurable cost” means the point estimates favour the frozen arms with intervals

including zero, not a demonstrated equivalence. Freezing is implemented inside this Mamba-3 layer; a conclusion about input-dependence in general would need other blocks and tasks.

10. **Gate interventions.** The gate-to-additive substitution has been made in both hosts and both directions. In the Mamba-3 block the verdict is *not rescued* on both transitions, with upper interval bounds of 0.486 and 0.362 against a threshold of 0.5 (the first close to it). In S4D the width-matched arms are larger than the donor block, and on the reverse transition the gate carries a small increment over the additive branch that it does not carry on the original transition; this direction dependence is unresolved.

11. **Strength of the pre-specification evidence.** Thresholds were pre-specified internally; for all but the last three follow-up grids the supporting evidence is local file times and run-directory names, not an external time-stamped registry (Section 3.5). The process-level repeat check is a single observation.

12. **Construction checks.** For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.

13. **Bearing identity.** All XJTU-SY transitions and the replicating Paderborn setting share bearings between training and evaluation, and in each a bearing’s identity determines its class. On Paderborn bearings disjoint from the training set, every examined architecture, S4D included, transferred close to the majority-class level and none of the contrasts replicated (Section 4.8). The conclusions are therefore limited to cross-condition drift with shared bearing identity; robustness to unseen bearings is not supported by these data.

14. **Sample size.** Every cell mean uses five seeds. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem, A parameterisation, gateless versus S4D) are not resolved at this size; conclusions rest on effect sizes that clear the pre-specified thresholds, not on significance tests, which were never adjudication criteria.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

794 7. Conclusion

795 In this Mamba-3 block, on one controlled drift $\times$ noise testbed with
796 matched-noise training, freezing the input-dependent Δ , A , B , C projections
797 costs at most about 4 pp (an estimate that five seeds leave imprecise) while
798 15–20 pp of advantage over a time-invariant SSM remains, and freezing the
799 trapezoidal weight and rotation angles as well—leaving no input-dependent
800 tensor in the scan—costs nothing measurable and leaves 21–24 pp of ad-
801 vantage with fewer parameters. The multiplicative output gate is the one
802 ablated component whose removal causes a large, one-signed loss (9–23 pp,
803 returning the block to roughly the level of plain S4D), yet adding the gate
804 to S4D recovers no more than the added structure does: 17–20% of the gap
805 for a narrow graft, about 30% for one built in the donor’s width, and at least
806 as much for an additive branch of the same size. Conversely, an additive
807 branch of identical size and initial weights, placed where the gate was in
808 the Mamba-3 block, recovers none of the loss on either transition. The gate
809 is thus necessary as a multiplicative operator in its native host, and what
810 transfers to a time-invariant host is added structure rather than gating; the
811 gate-related differences concentrate on the minority class. The first two steps
812 replicate on the reverse transition and on Paderborn bearings when bearing
813 identity is shared, but not on disjoint Paderborn bearings, where no exam-
814 ined architecture transfers. The evidence rests on cross-condition drift with
815 shared bearing identity, five seeds, and final-epoch reporting under a fixed
816 budget, with the earlier evaluation-condition epoch selection retained as a
817 sensitivity analysis. We offer the three-step chain—freeze, remove, add—as
818 a protocol for turning architecture comparisons into component-level tests,
819 with the caveat that the interventions must be matched in width and capac-
820 ity for the add step to be conclusive, and we release our adjudication trail,
821 including the incidents in which automated adjudication departed from the
822 pre-specified criteria.

823 CRediT authorship contribution statement

824 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
825 analysis, Investigation, Data curation, Writing – original draft, Writing –
826 review & editing, Visualization, Supervision, Project administration.

827 Declaration of competing interest

828 The authors declare that they have no known competing financial inter-
829 ests or personal relationships that could have appeared to influence the work
830 reported in this paper.

831 Declaration of Generative AI and AI-assisted technologies in the 832 writing process

833 During the preparation of this work the authors used LLM-based agents
834 to execute and cross-check pre-specified analysis and adjudication passes as
835 described in Appendix C, and to assist with manuscript drafting and figure
836 generation. After using these tools, the authors reviewed and edited the con-
837 tent as needed and take full responsibility for the content of the published
838 article. All numeric thresholds, adjudication rulings, and scientific interpre-
839 tations reported in this paper were fixed and made by the human authors,
840 as detailed in Appendix C.

841 Data availability

842 The code, pre-specification artifacts, and adjudication trail for this study
843 are available at <https://github.com/jeff-1980/bm3-jizi>. The repository
844 includes all follow-up grids (435 cells with per-epoch curves), their pre-
845 specifications, drivers, construction checks and decision memos and the har-
846 ness’s own imported modules, so the training entry point runs from a fresh
847 checkout.

848 Appendix A. Fairness protocol

849 *Appendix A.1. Evaluation-set fingerprint*

850 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
851 256 hash of the raw held-out test-label byte array, computed once per run
852 and asserted identical across every arm/condition/seed cell within a grid.
853 It fingerprints *that the same held-out label vector was scored every time*, not
854 model weights or predictions. It is also the split on which the best-epoch rule
855 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
856 five grids is:

857 6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de

858 This value is recomputed independently at adjudication time (not merely
859 copied from the run log) and cross-checked row-by-row against every grid’s
860 fairness report before any comparison across arms is treated as valid.

861 *Appendix A.2. Per-arm parameter counts*

862 Table A.12 lists every arm’s parameter count, each obtained directly via
863 `sum(p.numel() for p in model.parameters())` on the instantiated model
864 (not estimated), cross-verified against an independently maintained archi-
865 tecture map.

Table A.12: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

866 *Appendix A.3. Five-batch pre-specification timeline*

867 Table A.13 lists, for each of the five pre-specified grids, the artifact that
868 fixed its adjudication thresholds, that artifact’s creation/modification date,
869 and the corresponding grid’s run-directory start time. In every case the
870 prereg artifact predates the grid it governs.

871 Two honest caveats. First, the baseline batch predates the orchestration
872 infrastructure used for the other four batches by five days; its prereg trail
873 rests on the task card’s creation timestamp alone, not on the same automat-
874 ed executor/reviewer machinery documented in [Appendix C](#), and we state
875 this rather than implying a uniform process across all five batches. Second,
876 for the component-ablation grid, two additional JSON files bearing “pre-
877 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
878 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
879 time; both are explicitly marked void in the corresponding decision artifact’s

Table A.13: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (m- time 18:26:54)	2026-07-02	2026-07-02 18:27
Component ab- lation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

880 `void_preregs` field, and neither was used to adjudicate any result in this pa-
881 per. They are the on-disk residue of the first post-hoc criterion substitution
882 described in [Appendix C](#), retained here (rather than deleted) as part of the
883 released adjudication trail.

884 Appendix B. Confusion matrices and a repeat-run check

885 Appendix B.1. Full confusion-matrix set

886 Table [B.14](#) gives the complete confusion-matrix set underlying the per-
887 class analysis of [Section 4.10](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
888 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
889 counts were dumped per-sample and re-verified against the reported recal-
890 l/precision/F1 metrics cell-by-cell before use.

Table B.14: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
BM3-frozen	−6 dB	16,613	907	1,664	7,056
	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
S4D+gate	−6 dB	15,634	1,886	2,543	6,177
	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
frozen−gate	−6 dB	13,267	4,253	3,244	5,476
	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
S4D	−6 dB	13,200	4,320	5,469	3,251
	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

891 *Appendix B.2. Repeat-run check*

892 Because two independent process runs of an identical configuration are
893 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
894 ductions in the selective-scan and normalisation ops), we measured a process-
895 level noise floor directly: the same arm, condition, and seed (`bm3.frozen`,
896 clean, seed 0, 2 epochs) was run twice in separate processes with identical
897 evaluation fingerprints and identical parameter counts confirmed. The two
898 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

899 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
900 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
901 run, and therefore neither an estimate nor a bound of process variability for
902 full-length runs. It is an order of magnitude below the seed-level variabil-
903 ity observed within the deep-noise band across the curve tables underlying
904 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
905 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
906 are judged against seed-level variability, not against this single repeat.

907 **Appendix C. Automation disclosure and adjudication gates**

908 *Appendix C.1. Automation disclosure*

909 The mechanical stages of the analysis pipeline—applying pre-specified nu-
910 meric thresholds to aggregated results, re-computing aggregates, re-verifying
911 evaluation fingerprints, and checking each decision against its pre-specified
912 rule—were carried out by LLM-based coding agents under an orchestration
913 harness, with every step auditable in the released trail. Manuscript drafting
914 and figure generation were also assisted by such agents (main text, Declara-
915 tion of Generative AI).

916 *Appendix C.2. Incidents*

917 Three incidents occurred in the original grids, each caught before an er-
918 roneous artefact entered a reported number: (1) on the frozen-selectivity
919 grid an automated decision substituted an unspecified criterion for the pre-
920 specified 10 pp threshold; re-applying the original rule gives the reported out-
921 come, “not the primary mechanism”; (2) on the component-ablation grid an
922 automated pass substituted an interval heuristic for the pre-specified thresh-
923 old, and post-hoc rule files were generated; the true pre-specification predated
924 the grid and was re-applied, and the post-hoc files are retained and marked
925 void; (3) on the graft grid a construction stage was falsely marked as passed;
926 the grid was re-executed from scratch under a corrected task card. Devia-
927 tions in the later follow-up grids (for example a bootstrap made deterministic
928 after first use, and rounding corrections) are recorded in the repository’s de-
929 cision memos. Full descriptions, with dates and artefact names, are given in
930 Supplementary Material S1.

931 *Appendix C.3. Human adjudication gates*

932 All numeric adjudication thresholds, every ruling on an incident, the de-
933 cision to report the incidents, and every interpretive claim in Section 5 were
934 fixed or made by the human authors. No agent could overwrite a protected
935 decision artefact, alter a pre-specified threshold or mark its own output as
936 final.

937 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

938 In line with Elsevier’s policy, AI tools are not listed as authors and the
939 human authors take full responsibility for the content. We extend the dis-
940 closure beyond the writing process to the adjudication pipeline because that
941 is where automation could affect the reported conclusions.

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